

David Munoz Tord

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Competencies

R	10 yrs
Python	5 yrs
Git/Bash	10 yrs
App Dev (Shiny)	5 yrs

Libraries & Tools

- Reproducibility: Git, Docker
- Reporting: Quarto, R Markdown
- Viz: ggplot2, plotly, echarts4r
- AI/ML: rstan, brms, PyTorch
- Data: data.table, SQL, API

Domains

- Biomedical Data Science
- High-Dimensional Analysis
- Reproducible Pipelines
- Statistical Consulting
- Open Science & Mentoring

Languages

French	Native
English	Fluent
Spanish	Native
Italian	Basic
Japanese	Basic

Profile

Data Scientist translating complex high-dimensional biomedical data into actionable insights. Expert in building reproducible analytical pipelines (R, Docker, GitHub, Quarto) and open-source tools. Proven track record of guiding researchers from experimental design to publication-ready figures and code sharing.

Experience

Senior R Developer — **02/2025 - Present**

Cytel for Johnson & Johnson

– Advise and consult research laboratories on the selection of statistical methodologies and technology stacks for high-dimensional biomedical data analysis, particularly for clinical efficacy and oncology endpoints

– Develop, document, and maintain reproducible reporting pipelines (Quarto, R Markdown) deployed via containerized environments (Docker/CI-CD)

– Build reusable, open-source R packages to standardize and accelerate data processing workflows across multiple research teams

– Design, performance testing, and maintenance of analytical infrastructures (Posit Server) hosting applications in a strictly regulated environment

Data Science Consultant — **01/2024 - 02/2025**

Freelance - Pharmaceutical & Tech

– Led end-to-end biomedical data projects, translating researcher requirements into scalable architecture and reproducible deployment

– Developed high-performance custom R containerized apps integrating rigorous unit testing: See here for the Chemical Intelligence Platform

Data Scientist — **03/2023 - 01/2024**

FlowBank

– Architected and deployed a comprehensive data science environment for the internal data team, standardizing reproducible workflows (Docker, Git) and analytical tools

– Analyzed high-frequency datasets to develop and maintain machine learning pipelines for trade anomaly and fraud detection

– Developed custom internal R packages and containerized Shiny applications, enabling business teams to interactively explore complex multivariate data

Key Contributor & Developer — **03/2023 - Present**

Opifex

– **echarts4r (R and JavaScript):** Lead maintainer, high-impact visualization package. Widely adopted across the R community for generating complex, interactive charts for research groups.

– **fiRebase (R and JavaScript):** Lead maintainer, enabling secure backend integrations for interactive web applications.

PhD Candidate in Computational Neuroscience — **03/2020 - 01/2022**

Novo Nordisk & University of Geneva

– Architected an HPC cluster (Slurm) to optimize parallel analytical workflows on bare hardware

– Engineered Linux servers and storage architecture to securely process high-dimensional biomedical data

– Provided statistical support for experimental design and hypothesis validation

– Analyzed high-dimensional biological datasets (fMRI/ERP) via signal processing, sparse matrices, and clustering

– Ensured research reproducibility by maintaining traceable analytical pipelines (Git, Docker, R Markdown)

– Supported peer-reviewed publications with high-quality data visualizations (ggplot2) and open-source code

Education

Ph.D. Candidate in Computational Neuroscience

03/2019 - 01/2022
University of Geneva
Focus: Large-scale MRI data processing & Bayesian modeling

MSc in Computational Neuroscience

09/2018 - 03/2020
University of Geneva
Thesis: High-resolution fMRI signals using tICA

Complementary Studies in Computer Science

09/2017 - 06/2018
UMass Amherst - USA
Capstone: Machine Learning for neuroimaging data

BSc in Cognitive Sciences

09/2014 - 06/2017
University of Geneva
Thesis: Early detection of ASD event-related potentials

Extracurricular

- Top 12.5% on GitHub (2026)
- Top 2% on StackOverflow (2024)
- Global Fellow scholarship (2017)
- Co-founder of Rocket-Courrier, an eco-responsible bike messenger cooperative.

Resume Source

- Quarto code

Asssistant Lecturer in Biostatistics — 09/2020 - 01/2022
UniGe - Neuroscience Master

– Taught introductory biostatistics and led practical workshops in R

– Created and maintained course materials

Open Source Projects, Training and Community

See my GitHub profile for a more comprehensive list.

Author & Lead Developer — Packages
R, Python, Shiny

– Participated in the R Project Sprint 2026 and contributed to base R development alongside **R Core** contributors.

– **RConsortium member**: Lead contributor to the pharmaverse framework (rtables, tern, junco, formatters), supporting reproducible clinical reporting.

– **circle (rOpenSci)**: Official maintainer of the CircleCI DevOps integration package for R, promoting CI/CD in academic and enterprise research.

– **Major contributions**: Contributed source code to reference ecosystem packages, notably **ggplot2**, **brms**, **AFNI**.

Co-President — We Data
Training, Vulgarisation, Data Science

– Organized training, hackathons, and UniGe "R-Lunches" to share methods.

– Grew our YouTube channel about statistics to 3,500+ followers.

Publications

See my publications page for more.

Differential contributions of ventral striatum subregions to the motivational and hedonic components of the affective processing of reward. Pool, E. R., Munoz Tord, D. , Delplanque, S., Stussi, Y., Cereghetti, D., Vuilleumier, P., & Sander, D. *Journal of Neuroscience* (2022) - Paper

Effect of GLP-1 receptor agonist on self-reported liking scales and its neural responses to food stimuli. Coppin, G., Munoz Tord, D., Pool, Cereghetti, D., Golay, A., Sander, D., & Pataky, Z. *Journal of Obesity* (2023) - Paper

3D-printed pacifier-shaped mouthpiece for fMRI-compatible gustometers. Munoz Tord, D., Coppin, G., Pool, E. R., Mermoud, C., Pataky, Z., Sander, D., & Delplanque, S. *Eneuro* (2021) - Paper

Early spatial attention deployment towards aggressive voices. Burra, N., Kerzel, D., Munoz Tord, D., Grandjean, D., & Ceravolo, L. *Social Cognitive and Affective Neuroscience* (2019) - Paper

Professional References

Gabriel Becker, **Statistical Computing Consultant and R-core contributor**

Pierre Saouter, **Head Data Scientist at FlowBank**

Ben Meuleman, **Statistician at University of Geneva**

Jose Nunes, **Professor of Biostatistics at the University of Geneva**

Eva R. Pool, **Professor at University of Geneva**